

Question number	Answer	Mark																								
3(a)	<table><tr><th rowspan="2">Statement</th><th colspan="4">Type of artificial immunity</th></tr><tr><th>both active and passive</th><th>active only</th><th>passive only</th><th>neither active nor passive</th></tr><tr><td>Antibodies are injected into the person</td><td></td><td></td><td>X</td><td></td></tr><tr><td>B cells differentiate into plasma cells</td><td></td><td>X</td><td></td><td></td></tr><tr><td>Memory cells are formed</td><td></td><td>X</td><td></td><td></td></tr></table>	Statement	Type of artificial immunity				both active and passive	active only	passive only	neither active nor passive	Antibodies are injected into the person			X		B cells differentiate into plasma cells		X			Memory cells are formed		X			(3)
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3(b)(i)	A description that includes the following points: - engulf the virus / phagocytosis of the virus (1) - digestion of the virus (1) - antigen presentation to {T helper / CD4} {cells / lymphocytes} (1)	ACCEPT pathogen, if in context of virus reference to a macrophage being a phagocyte, if in context of viruses DO NOT ACCEPT kills the virus ACCEPT macrophage is an {antigen presenting cell / APC} to T helper cells DO NOT ACCEPT to T killer cells	(3)

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3(b)(ii)	An explanation that includes four of the following points: • T helper cells {activate / stimulate} B cells (to divide) (1) • because antibody will be needed for opsonisation (1) • T helper cells {activate / stimulate} T killer cells (to divide) (1) • because T killer cells destroy (virus-infected) cells (1) • so that virus can be {engulfed / destroyed} by macrophages (1)	ACCEPT stimulate humoral response ACCEPT antibodies prevent viruses binding to host cells ACCEPT stimulate cell-mediated response ACCEPT a description DO NOT ACCEPT killed pathogen in context of bacteria	(4)